



Enhancing Policyholder Data Accuracy in the Insurance Industry with ML-Based Deduplication

Case Study



Estimated Annual Business Impact

\$900K+

Annual savings from reduced analyst effort, duplicate communication, and fraud-related inefficiencies.

20-30%

Faster quote-to-bind turnaround due to clean policyholder records.

70%

Reduction in manual effort in data cleansing with automation





Business Problem

In the insurance industry, data quality is foundational to underwriting, claims processing, and regulatory compliance. Yet, many carriers face persistent challenges with duplicate and inconsistent policyholder records:



8–15% of customer records across quoting, underwriting, and claims systems are duplicates.



Traditional rule-based deduplication misses ~35% of potential matches due to misspellings, inconsistent formats, and nickname usage.



Manual efforts to clean data cost 80–120 analyst hours per month, delaying quote generation and claims adjudication.



Poor data quality leads to duplicate mailings, inaccurate risk assessments, and inflated loss ratios.

These issues increase operational cost, regulatory exposure, and customer dissatisfaction.

High-Level Solution

ML-BASED POLICYHOLDER MATCHING

- Used fuzzy matching (Levenshtein, Jaro-Winkler) across fields like name, date of birth, address, policy number, and SSN.
- Applied ML models to detect likely duplicates even with typographical and format differences.

HUMAN-IN-THE-LOOP ACTIVE LEARNING

- Focused analyst review on only uncertain cases, reducing manual validation workload by over 65%.
- Continuously improved the matching model using validated training data.

GOLDEN RECORD CONSOLIDATION

Unified duplicate policyholder records into a single profile for consistent underwriting, claims, and servicing.

TECHNOLOGY STACK

- Python + Dedupe: Deduplication engine.
- Azure SQL + Data Factory: Data transformation and pipeline orchestration.
- Prefect: Scheduled and event-driven deduplication workflows.
- Streamlit: Internal UI for data quality and validation review.
- Guidewire + Salesforce: Integration with policy administration and customer systems.

Insurance

Impact Metrics

Impact Area	Before	After	Business Impact
Duplicate Policyholder Records	8–15%	<2%	↑ Reliable customer identity across systems
Manual Data Cleansing Time	80–120 hrs/month	<35 hrs/month	↓ 60–70% manual effort
Claims Duplication Issues	Frequent	Rare	↓ False positives in claims and fraud triggers
Underwriting Inconsistencies	Common due to data silos	Reduced	↑ Consistent risk scoring
Customer Communications Overlap	~20%	<5%	↓ Redundant outreach and mailing costs
Regulatory Data Accuracy	Moderate	High	↑ Audit readiness and compliance posture



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